

SQL Practice Notes

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Problem 1: Unique ID

Table: Employees

Column Name	Type
id	int
name	varchar

id is the primary key (column with unique values) for this table.

Each row of this table contains the id and the name of an employee in a company.

Table: EmployeeUNI

Column Name	Type
id	int
unique_id	int

(id, unique_id) is the primary key (combination of columns with unique values) for this table.

Each row of this table contains the id and the corresponding unique id of an employee in the company.

Write a solution to show the unique ID of each user, If a user does not have a unique ID replace just show null.

Return the result table in any order.

The result format is in the following example.

Example 1:

Input:

Employees table:

id	name
1	Alice
7	Bob
11	Meir
90	Winston
3	Jonathan

EmployeeUNI table:

```

+-----+-----+
| id | unique_id |
+-----+-----+
| 3 | 1 |
| 11 | 2 |
| 90 | 3 |
+-----+-----+

```

Output:

```

+-----+-----+
| unique_id | name |
+-----+-----+
| null | Alice |
| null | Bob |
| 2 | Meir |
| 3 | Winston |
| 1 | Jonathan |
+-----+-----+

```

```

# Write your MySQL query statement below
SELECT
    eu.unique_id,
    e.name
FROM Employees e
LEFT JOIN EmployeeUNI eu
ON e.id = eu.id;

```

Problem 2: Sales Report

Table: Sales

```

+-----+-----+
| Column Name | Type |
+-----+-----+
| sale_id | int |
| product_id | int |
| year | int |
| quantity | int |
| price | int |
+-----+-----+

```

(sale_id, year) is the primary key (combination of columns with unique values) of this table.

product_id is a foreign key (reference column) to Product table.

Each row of this table shows a sale on the product product_id in a certain year.

Note that the price is per unit.

Table: Product

```

+-----+-----+
| Column Name | Type |
+-----+-----+

```

```
+-----+-----+
| product_id | int      |
| product_name | varchar |
+-----+-----+
```

product_id is the primary key (column with unique values) of this table.
Each row of this table indicates the product name of each product.

Write a solution to report the product_name, year, and price for each sale_id in the Sales table.
Return the resulting table in any order.

```
SELECT
    p.product_name,
    s.year,
    s.price
FROM Sales s
JOIN Product p
ON s.product_id = p.product_id;
```

Problem 3: Customer Visits without Transactions

Table: Visits

```
+-----+-----+
| Column Name | Type      |
+-----+-----+
| visit_id    | int       |
| customer_id | int       |
+-----+-----+
```

visit_id is the column with unique values for this table.
This table contains information about the customers who visited the mall.

Table: Transactions

```
+-----+-----+
| Column Name | Type      |
+-----+-----+
| transaction_id | int       |
| visit_id      | int       |
| amount        | int       |
+-----+-----+
```

transaction_id is column with unique values for this table.
This table contains information about the transactions made during the visit_id.

```

SELECT
    v.customer_id,
    COUNT(*) AS count_no_trans
FROM Visits v
WHERE NOT EXISTS
(
    SELECT 1
    FROM Transactions t
    WHERE t.visit_id = v.visit_id
)
GROUP BY v.customer_id;

```

Problem 4: Average Processing Time

Table: Activity

Column Name	Type
machine_id	int
process_id	int
activity_type	enum
timestamp	float

The table shows the user activities for a factory website.

(machine_id, process_id, activity_type) is the primary key (combination of columns with unique values) of this table.

machine_id is the ID of a machine.

process_id is the ID of a process running on the machine with ID machine_id.

activity_type is an ENUM (category) of type ('start', 'end').

timestamp is a float representing the current time in seconds.

'start' means the machine starts the process at the given timestamp and 'end' means the machine ends the process at the given timestamp.

The 'start' timestamp will always be less than or equal to the 'end' timestamp for every (machine_id, process_id) pair.

It is guaranteed that each (machine_id, process_id) pair has a 'start' and 'end' timestamp.

There is a factory website that has several machines each running the same number of processes. Write a solution to find the average time each machine takes to complete a process.

The time to complete a process is the 'end' timestamp minus the 'start' timestamp.

The average time is calculated by the total time to complete every process on the machine divided by the number of processes that were run.

The resulting table should have the machine_id along with the average time as processing_time, which should be rounded to 3 decimal places.

Return the result table in any order.

```

SELECT
    a.machine_id,
    ROUND(AVG(b.timestamp - a.timestamp), 3) AS processing_time
FROM Activity a
JOIN Activity b
ON a.machine_id = b.machine_id
AND a.process_id = b.process_id
WHERE a.activity_type = 'start'
AND b.activity_type = 'end'
GROUP BY a.machine_id;

```

Problem 5: Employee Bonus

Table: Employee

Column Name	Type
empId	int
name	varchar
supervisor	int
salary	int

empId is the column with unique values for this table.

Each row of this table indicates the name and the ID of an employee in addition to their salary and the id of their manager.

Table: Bonus

Column Name	Type
empId	int
bonus	int

empId is the column of unique values for this table.

empId is a foreign key (reference column) to empId from the Employee table.

Each row of this table contains the id of an employee and their respective bonus.

Write a solution to report the name and bonus amount of each employee who satisfies either of the following:

The employee has a bonus less than 1000.

The employee did not get any bonus.

```

SELECT
    e.name,
    b.bonus
FROM Employee e
LEFT JOIN Bonus b
ON e.empId = b.empId
WHERE b.bonus < 1000
OR b.bonus IS NULL;

```

Problem 6: Student Examinations

Table: Students

Column Name	Type
student_id	int
student_name	varchar

student_id is the primary key (column with unique values) for this table.
Each row of this table contains the ID and the name of one student in the school.

Table: Subjects

Column Name	Type
subject_name	varchar

subject_name is the primary key (column with unique values) for this table.
Each row of this table contains the name of one subject in the school.

Table: Examinations

Column Name	Type
student_id	int
subject_name	varchar

There is no primary key (column with unique values) for this table. It may contain duplicates.

Each student from the Students table takes every course from the Subjects table.
Each row of this table indicates that a student with ID student_id attended the exam of subject_name.

Write a solution to find the number of times each student attended each exam.
Return the result table ordered by student_id and subject_name.
The result format is in the following example.

We first need every possible (student, subject) pair. Using CROSS JOIN. Then, attach the examination records using LEFT JOIN.

```
SELECT
    s.student_id,
    s.student_name,
    sub.subject_name,
    COUNT(e.subject_name) AS attended_exams
FROM Students s
CROSS JOIN Subjects sub
LEFT JOIN Examinations e
ON s.student_id = e.student_id
AND sub.subject_name = e.subject_name
GROUP BY
    s.student_id,
    s.student_name,
    sub.subject_name
ORDER BY
    s.student_id,
    sub.subject_name;
```

Problem 7: Immediate Food Delivery

Table: Delivery

Column Name	Type
delivery_id	int
customer_id	int
order_date	date
customer_pref_delivery_date	date

delivery_id is the column of unique values of this table.

The table holds information about food delivery to customers that make orders at some date and specify a preferred delivery date (on the same order date or after it).

If the customer's preferred delivery date is the same as the order date, then the order is called immediate; otherwise, it is called scheduled.

The first order of a customer is the order with the earliest order date that the customer made. It is guaranteed that a customer has precisely one first order.

Write a solution to find the percentage of immediate orders in the first orders of all customers, rounded to 2 decimal places.

```

SELECT
    ROUND(
        AVG(order_date = customer_pref_delivery_date) * 100,
        2
    ) AS immediate_percentage
FROM Delivery
WHERE (customer_id, order_date) IN
(
    SELECT
        customer_id,
        MIN(order_date)
    FROM Delivery
    GROUP BY customer_id
);

```

Problem 8: Managers and Reports

Table: Employees

Column Name	Type
employee_id	int
name	varchar
reports_to	int
age	int

employee_id is the column with unique values for this table.

This table contains information about the employees and the id of the manager they report to. Some employees do not report to anyone (reports_to is null).

For this problem, we will consider a manager an employee who has at least 1 other employee reporting to them.

Write a solution to report the ids and the names of all managers, the number of employees who report directly to them, and the average age of the reports rounded to the nearest integer.

Return the result table ordered by employee_id.

The result format is in the following example.

Example 1:

Input:

Employees table:

employee_id	name	reports_to	age
9	Hercy	null	43
6	Alice	9	41
4	Bob	9	36

2	Winston	null	37
---	---------	------	----

Output:

employee_id	name	reports_count	average_age
9	Hercy	2	39

Explanation: Hercy has 2 people report directly to him, Alice and Bob. Their average age is $(41+36)/2 = 38.5$, which is 39 after rounding it to the nearest integer.

```
SELECT
    m.employee_id,
    m.name,
    COUNT(e.employee_id) AS reports_count,
    ROUND(AVG(e.age)) AS average_age
FROM Employees m
JOIN Employees e
ON m.employee_id = e.reports_to
GROUP BY
    m.employee_id,
    m.name
ORDER BY
    m.employee_id;
```

Problem 9: Primary Department

Table: Employee

Column Name	Type
employee_id	int
department_id	int
primary_flag	varchar

(employee_id, department_id) is the primary key (combination of columns with unique values) for this table.

employee_id is the id of the employee.

department_id is the id of the department to which the employee belongs.

primary_flag is an ENUM (category) of type ('Y', 'N'). If the flag is 'Y', the department is the primary department for the employee. If the flag is 'N', the department is not the primary.

Employees can belong to multiple departments. When the employee joins other departments, they need to decide which department is their primary department. Note that when an employee belongs to only one department, their primary column is 'N'.

Write a solution to report all the employees with their primary department. For employees who belong to one department, report their only department. Return the result table in any order. The result format is in the following example.

Example 1:

Input:

Employee table:

employee_id	department_id	primary_flag
1	1	N
2	1	Y
2	2	N
3	3	N
4	2	N
4	3	Y
4	4	N

Output:

employee_id	department_id
1	1
2	1
3	3
4	3

```
SELECT
    e.employee_id,
    e.department_id
FROM Employee e
JOIN
(
    SELECT
        employee_id,
        COUNT(*) AS cnt
    FROM Employee
    GROUP BY employee_id
) t
ON e.employee_id = t.employee_id
WHERE t.cnt = 1
OR e.primary_flag = 'Y';
```

Problem 10: Consecutive Numbers

Table: Logs

+-----+-----+		
Column Name	Type	
+-----+-----+		
id	int	
num	varchar	
+-----+-----+		

In SQL, id is the primary key for this table.
id is an autoincrement column starting from 1.

Find all numbers that appear at least three times consecutively.
Return the result table in any order.
The result format is in the following example.

Example 1:

Input:

Logs table:

+----+----+		
id	num	
+----+----+		
1	1	
2	1	
3	1	
4	2	
5	1	
6	2	
7	2	
+----+----+		

Output:

+-----+	
ConsecutiveNums	
+-----+	
1	
+-----+	

```
SELECT DISTINCT
    l1.num AS ConsecutiveNums
FROM Logs l1
JOIN Logs l2
ON l2.id = l1.id + 1
JOIN Logs l3
ON l3.id = l1.id + 2
WHERE l1.num = l2.num
AND l2.num = l3.num;
```

Problem 11: Product Price at a Given Date

Table: Products

Column Name	Type	
product_id	int	
new_price	int	
change_date	date	

(product_id, change_date) is the primary key (combination of columns with unique values) of this table.

Each row of this table indicates that the price of some product was changed to a new price at some date.

Initially, all products have price 10.

Write a solution to find the prices of all products on the date 2019-08-16.

Return the result table in any order.

The result format is in the following example.

Example 1:

Input:

Products table:

product_id	new_price	change_date
1	20	2019-08-14
2	50	2019-08-14
1	30	2019-08-15
1	35	2019-08-16
2	65	2019-08-17
3	20	2019-08-18

Output:

product_id	price
2	50
1	35
3	10

```

SELECT
    p.product_id,
    p.new_price AS price
FROM Products p
JOIN
(
    SELECT
        product_id,
        MAX(change_date) AS latest_date
    FROM Products
    WHERE change_date <= '2019-08-16'
    GROUP BY product_id
) t
ON p.product_id = t.product_id
AND p.change_date = t.latest_date
UNION
SELECT
    product_id,
    10 AS price
FROM Products
GROUP BY product_id
HAVING MIN(change_date) > '2019-08-16';

```

Problem 12: Employees Whose Manager Left

Table: Employees

Column Name	Type
employee_id	int
name	varchar
manager_id	int
salary	int

In SQL, employee_id is the primary key for this table.

This table contains information about the employees, their salary, and the ID of their manager. Some employees do not have a manager (manager_id is null).

Find the IDs of the employees whose salary is strictly less than \$30000 and whose manager left the company. When a manager leaves the company, their information is deleted from the Employees table, but the reports still have their manager_id set to the manager that left.

Return the result table ordered by employee_id.

The result format is in the following example.

Example 1:

Input:

Employees table:

employee_id	name	manager_id	salary
3	Mila	9	60301
12	Antonella	null	31000
13	Emery	null	67084
1	Kalel	11	21241
9	Mikaela	null	50937
11	Joziah	6	28485

Output:

employee_id
11

```
SELECT
    employee_id
FROM Employees
WHERE salary < 30000
AND manager_id IS NOT NULL
AND manager_id NOT IN
(
    SELECT employee_id
    FROM Employees
)
ORDER BY employee_id;
```

Problem 13: High Earners in Department

Table: Employee

Column Name	Type
id	int
name	varchar
salary	int
departmentId	int

id is the primary key (column with unique values) for this table.

departmentId is a foreign key (reference column) of the ID from the Department table.

Each row of this table indicates the ID, name, and salary of an employee. It also contains the ID of their department.

Table: Department

+-----+-----+		
Column Name	Type	
+-----+-----+		
id	int	
name	varchar	
+-----+-----+		

id is the primary key (column with unique values) for this table.

Each row of this table indicates the ID of a department and its name.

A company's executives are interested in seeing who earns the most money in each of the company's departments. A high earner in a department is an employee who has a salary in the top three unique salaries for that department.

Write a solution to find the employees who are high earners in each of the departments.

Return the result table in any order.

The result format is in the following example.

Example 1:

Input:

Employee table:

+-----+-----+-----+-----+				
id	name	salary	departmentId	
+-----+-----+-----+-----+				
1	Joe	85000	1	
2	Henry	80000	2	
3	Sam	60000	2	
4	Max	90000	1	
5	Janet	69000	1	
6	Randy	85000	1	
7	Will	70000	1	
+-----+-----+-----+-----+				

Department table:

+-----+-----+		
id	name	
+-----+-----+		
1	IT	
2	Sales	
+-----+-----+		

Output:

+-----+-----+-----+			
Department	Employee	Salary	
+-----+-----+-----+			
IT	Max	90000	
IT	Joe	85000	
IT	Randy	85000	
IT	Will	70000	
Sales	Henry	80000	
Sales	Sam	60000	
+-----+-----+-----+			

```

SELECT
    d.name AS Department,
    e.name AS Employee,
    e.salary AS Salary
FROM Employee e
JOIN Department d
ON e.departmentId = d.id
WHERE
(
    SELECT COUNT(DISTINCT salary)
    FROM Employee
    WHERE departmentId = e.departmentId
    AND salary > e.salary
) < 3;

```

Problem 14: Delete Duplicate Emails

Table: Person

Column Name	Type
id	int
email	varchar

id is the primary key (column with unique values) for this table.

Each row of this table contains an email. The emails will not contain uppercase letters.

Write a solution to delete all duplicate emails, keeping only one unique email with the smallest id.

For SQL users, please note that you are supposed to write a DELETE statement and not a SELECT one.

```

DELETE p1
FROM Person p1
JOIN Person p2
ON p1.email = p2.email
AND p1.id > p2.id;

```

Problem 15: Products Sold Per Date

Table Activities:

+-----+-----+		
Column Name	Type	
+-----+-----+		
sell_date	date	
product	varchar	
+-----+-----+		

There is no primary key (column with unique values) for this table. It may contain duplicates.

Each row of this table contains the product name and the date it was sold in a market.

Write a solution to find for each date the number of different products sold and their names.

The sold products names for each date should be sorted lexicographically.

Return the result table ordered by sell_date.

The result format is in the following example.

Example 1:

Input:

Activities table:

+-----+-----+		
sell_date	product	
+-----+-----+		
2020-05-30	Headphone	
2020-06-01	Pencil	
2020-06-02	Mask	
2020-05-30	Basketball	
2020-06-01	Bible	
2020-06-02	Mask	
2020-05-30	T-Shirt	
+-----+-----+		

Output:

+-----+-----+-----+		
sell_date	num_sold	products
+-----+-----+-----+		
2020-05-30	3	Basketball,Headphone,T-shirt
2020-06-01	2	Bible,Pencil
2020-06-02	1	Mask
+-----+-----+-----+		

```
SELECT
    sell_date,
    COUNT(DISTINCT product) AS num_sold,
    GROUP_CONCAT(DISTINCT product ORDER BY product) AS products
FROM Activities
GROUP BY sell_date
ORDER BY sell_date;
```